DSA06 DATA HANDLING AND VISUALIZATION

Activity: Identify Seasonality Patterns | Slot D

Name: U ROSHITHA
Reg. No: 192425184
R. No: 38
Series: SP-D38 (Tourist arrivals, thousands)
Tool: R (ggplot2)

Creating the series

The code given in the question was run as it is, without any changes.

Code:

```
set.seed(51056)
m <- 12                  # observations per cycle (monthly data)
n <- m * 5               # 5 full cycles
t <- 1:n
trend <- 80.4 + 0.759 * t
arrivals <- round(trend + 19.7 * cos(2 * pi * (t - 5) / m) + rnorm(n, 0, 4.2), 1)
x <- ts(arrivals, frequency = m, start = c(2019, 1))
```

Output of print(x):

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2019	84.6	82.4	90.0	94.5	104.7	94.9	86.8	87.5	78.6	75.5	71.2	68.8
2020	82.8	91.4	97.3	106.1	112.6	111.4	104.2	93.2	89.8	83.8	77.7	80.5
2021	100.1	97.3	111.6	113.2	121.0	113.8	110.1	110.4	94.0	84.9	90.8	87.0
2022	97.0	116.8	113.6	132.0	134.5	128.0	130.0	110.4	102.6	92.6	97.6	100.5
2023	108.7	123.0	133.2	134.5	138.1	128.7	137.6	120.2	108.5	113.7	108.3	108.3

Task 1: Looking at the series

Code:

```
library(ggplot2)
cat("n =", length(x), "\n")
cat("Cycle length =", frequency(x), "\n")
cat("Number of full cycles =", length(x) / frequency(x), "\n")
fit <- lm(arrivals ~ t)      # t = 1:n
cat("Intercept =", round(coef(fit)[1], 3), "\n")
cat("Slope per month =", round(coef(fit)[2], 3), "\n")
df <- data.frame(t = t, arrivals = arrivals)
p1 <- ggplot(df, aes(t, arrivals)) +
  geom_line() +
  geom_smooth(method = "lm", se = FALSE, colour = "red", linetype = "dashed") +
  labs(title = "Monthly tourist arrivals (thousands)",
       x = "Month number (1 = Jan 2019)", y = "Arrivals (thousands)") +
  theme_bw(base_family = "Times New Roman", base_size = 12)
print(p1)
```

Note: in the lm() formula I used t (which is 1:n) because arrivals ~ 1:n gives an error in R, as n is also a variable.

Output:

```
n = 60
Cycle length = 12
Number of full cycles = 5
Intercept = 82.788
Slope per month = 0.675
```

Plot:

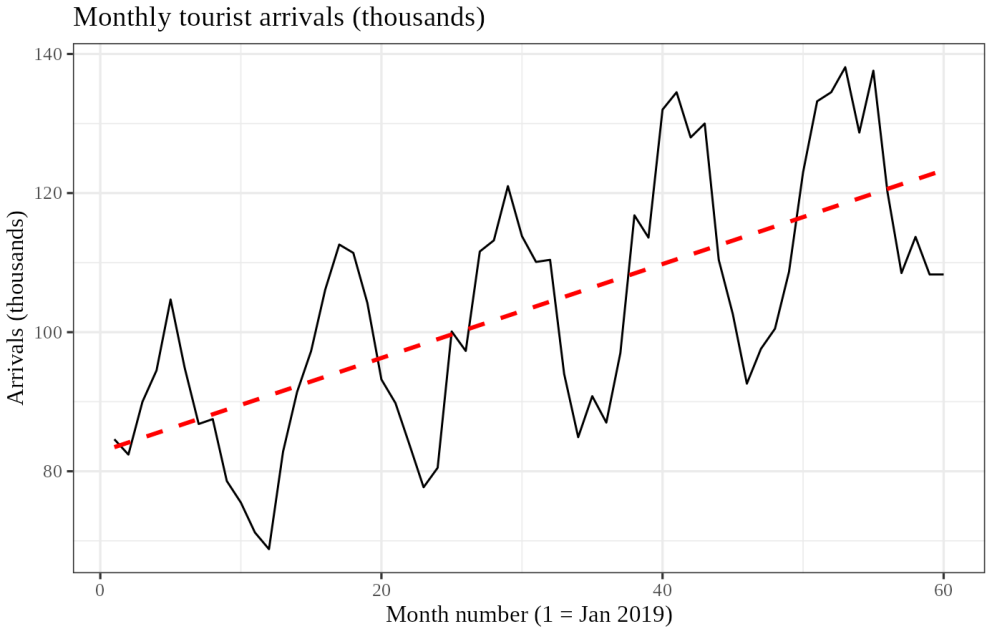


Figure 1: Time series plot of monthly tourist arrivals

Ans: n = 60 monthly observations (Jan 2019 to Dec 2023). The cycle length is 12 and there are 5 full cycles. The fitted slope is 0.675, that is arrivals increase by about 0.675 thousand (675 tourists) every month, so the overall trend is rising.

Task 2: The seasonal pattern

Code:

```
d <- decompose(x)
fig <- d$figure
names(fig) <- month.abb
print(round(rbind(figure = fig), 2))
cat("Peak month   :", names(which.max(fig)), round(max(fig), 2), "\n")
cat("Trough month :", names(which.min(fig)), round(min(fig), 2), "\n")
cat("Seasonal swing =", round(max(fig) - min(fig), 2), "\n")
sf <- data.frame(month = factor(month.abb, levels = month.abb), figure = as.numeric(fig))
p2 <- ggplot(sf, aes(month, figure)) +
  geom_col(fill = "grey60", colour = "black") +
  geom_hline(yintercept = 0) +
  labs(title = "Seasonal figure from decompose(x)", x = "Month", y = "Seasonal value") +
  theme_bw(base_family = "Times New Roman", base_size = 12)
print(p2)
```

Output:

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
figure	-6.53	2.57	8.72	15.53	19.85	12.98	8.85	0.78	-9.22	-17.14	-17.78	-18.6

```
Peak month   : May 19.85
Trough month : Dec -18.6
Seasonal swing = 38.45
```

Plot:

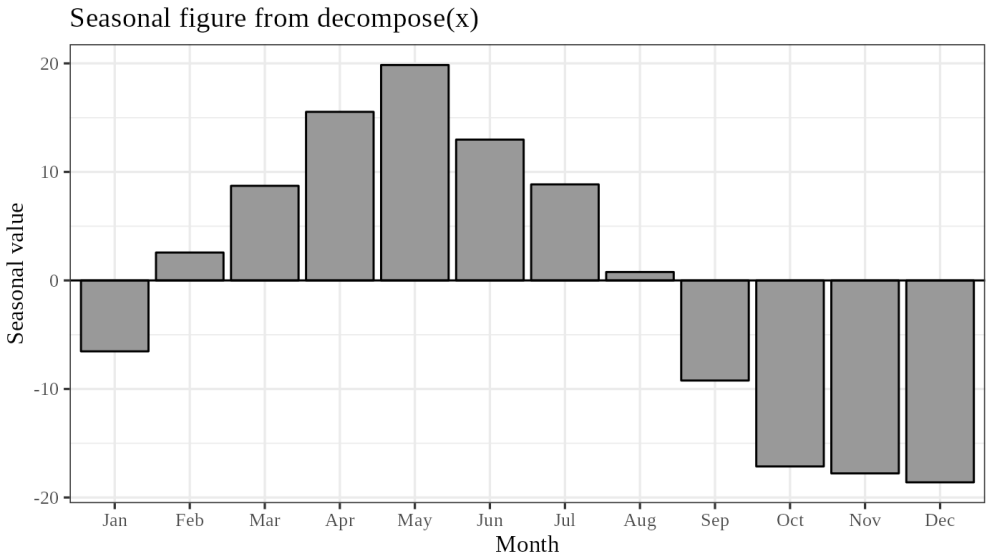


Figure 2: The 12 seasonal values

Ans: The peak month is May with a seasonal value of +19.85 and the trough month is December with a value of -18.60. The seasonal swing (peak minus trough) is 19.85 - (-18.60) = 38.45 thousand tourists.

Task 3: Is it consistent?

Code:

```
sp <- data.frame(arrivals = as.numeric(x),
                 cycle = factor(rep(1:5, each = 12)),
                 pos = rep(1:12, 5))
p3 <- ggplot(sp, aes(pos, arrivals, group = cycle, colour = cycle)) +
  geom_line() + geom_point() +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(title = "Seasonal plot: one line per cycle",
       x = "Position in cycle", y = "Arrivals (thousands)", colour = "Cycle") +
  theme_bw(base_family = "Times New Roman", base_size = 12)
print(p3)
mat <- matrix(arrivals, nrow = 5, byrow = TRUE,
              dimnames = list(paste0("C", 1:5), month.abb))
print(mat)
cat("Month with the highest value in each cycle:", month.abb[apply(mat, 1, which.max)], "\n")
cat("Cycles where the peak is in May =", sum(apply(mat, 1, which.max) == 5), "out of 5\n")
```

Output:

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
C1	84.6	82.4	90.0	94.5	104.7	94.9	86.8	87.5	78.6	75.5	71.2	68.8
C2	82.8	91.4	97.3	106.1	112.6	111.4	104.2	93.2	89.8	83.8	77.7	80.5
C3	100.1	97.3	111.6	113.2	121.0	113.8	110.1	110.4	94.0	84.9	90.8	87.0
C4	97.0	116.8	113.6	132.0	134.5	128.0	130.0	110.4	102.6	92.6	97.6	100.5
C5	108.7	123.0	133.2	134.5	138.1	128.7	137.6	120.2	108.5	113.7	108.3	108.3

Month with the highest value in each cycle: May May May May May
Cycles where the peak is in May = 5 out of 5

Plot:

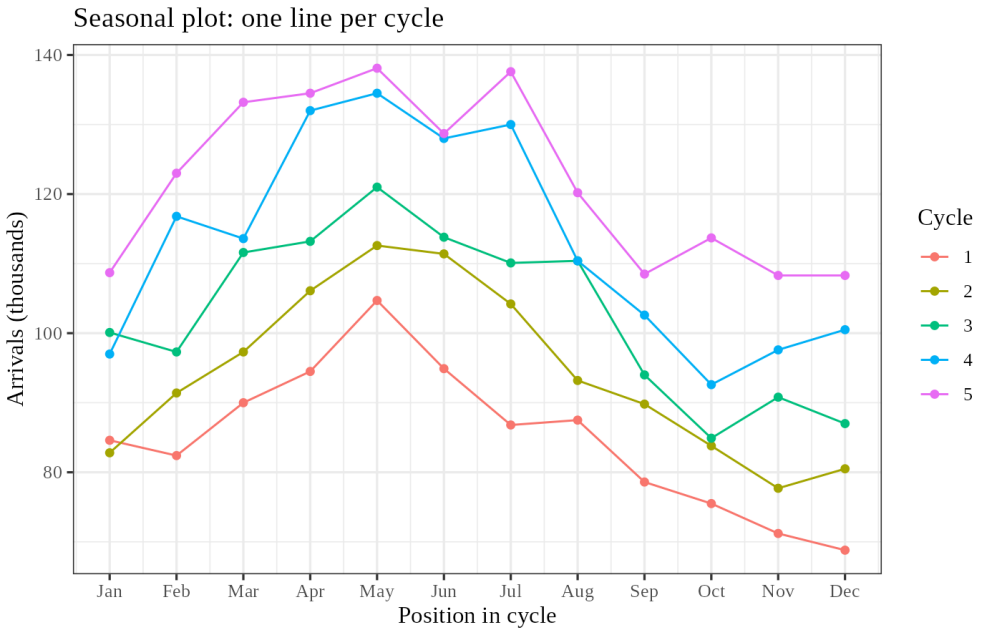


Figure 3: Seasonal plot, one line for each cycle

Ans: In all 5 out of 5 cycles the highest value of the year falls in May, the peak month from Task 2. So the pattern is consistent. All five lines have the same shape (rise to May, fall towards the end of the year); they only sit higher each year because of the rising trend.

Task 4: Confirming with autocorrelation

Code:

```
dd <- resid(lm(arrivals ~ t))
a <- acf(dd, lag.max = 12, plot = FALSE)
cat("ACF at lag 12 =", round(a$acf[13], 3), "\n")
cat("ACF at lag 6  =", round(a$acf[7], 3), "\n")
cat("Bound 2/sqrt(n) =", round(2 / sqrt(n), 3), "\n")
ac <- data.frame(lag = 1:12, acf = a$acf[2:13])
p4 <- ggplot(ac, aes(lag, acf)) +
  geom_col(width = 0.15, fill = "black") +
  geom_hline(yintercept = c(-1, 1) * 2 / sqrt(n), colour = "blue", linetype = "dashed") +
  geom_hline(yintercept = 0) +
  scale_x_continuous(breaks = 1:12) +
  labs(title = "ACF of the detrended series", x = "Lag (observations)", y = "ACF") +
  theme_bw(base_family = "Times New Roman", base_size = 12)
print(p4)
```

Output:

```
ACF at lag 12 = 0.74
ACF at lag 6  = -0.839
Bound 2/sqrt(n) = 0.258
```

Plot:

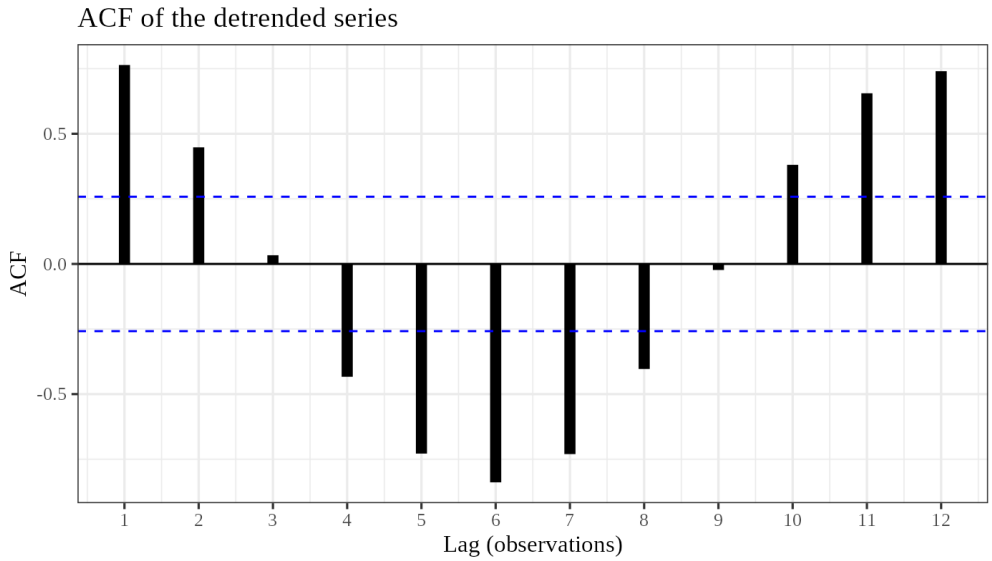


Figure 4: ACF of the detrended series (dashed lines = ± 2/sqrt(n))

Ans: The autocorrelation at lag 12 is 0.74 and at lag 6 it is -0.84. The significance bound is 2/sqrt(60) = 0.258. Both values are far outside the bound. The strong positive value at lag 12 means a month is similar to the same month one cycle (12 months) earlier. The strong negative value at lag 6 means a month is the opposite of the month half a cycle away (a high month is six months from a low month). Together they confirm a clear seasonal pattern repeating every 12 months.

Task 5: Additive or multiplicative?

Code:

```
range1 <- diff(range(arrivals[1:12]))
range5 <- diff(range(arrivals[49:60]))
cat("Range of first cycle =", round(range1, 1), "\n")
cat("Range of last cycle  =", round(range5, 1), "\n")
cat("Ratio (last / first) =", round(range5 / range1, 2), "\n")
da <- decompose(x, type = "additive")
adjusted <- x - da$seasonal
fit2 <- lm(as.numeric(adjusted) ~ t)
cat("Slope of adjusted series =", round(coef(fit2)[2], 3), "per month\n")
```

Output:

```
Range of first cycle = 35.9
Range of last cycle  = 29.8
Ratio (last / first) = 0.83
Slope of adjusted series = 0.77 per month
```

Ans: The range in the first cycle is 35.9 and in the last cycle it is 29.8, so the ratio (last / first) is 0.83. The swings did not grow even though the average level went up from about 85 to about 122 thousand. So the seasonality is additive (constant size swings, about 30 to 36 thousand), not multiplicative.

After removing the seasonal part with decompose(x, type = "additive") (subtracting the seasonal values), the slope of the adjusted series is 0.770 per month, which is close to the slope found in Task 1 (0.675).

Advice to management: Plan for a fixed yearly pattern, with about 38 thousand more tourists in May than in December, together with a steady rise of about 0.77 thousand arrivals every month, so staff, rooms and transport should be increased before April-May each year.